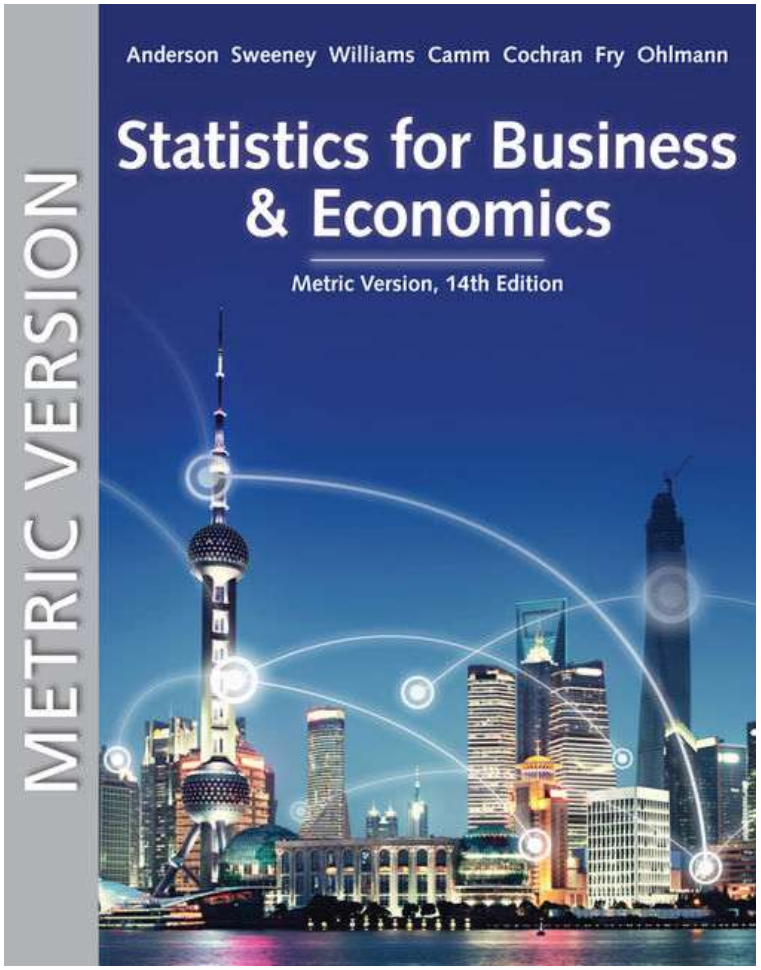
Statistics for Business and Economics (14e) Metric Version

Anderson, Sweeney, Williams, Camm, Cochran, Fry, Ohlmann

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Chapter 4 - Introduction to Probability



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Chapter 4 - Introduction to Probability

4.1 - Random Experiments, Counting Rules, and Assigning Probabilities

4.2 - Events and Their Probabilities

4.3 - Some Basic Relationships of Probability

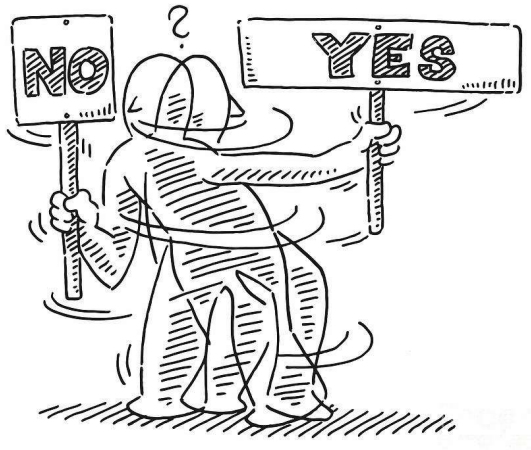
4.4 - Conditional Probability

4.5 - Bayes' Theorem

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Uncertainties

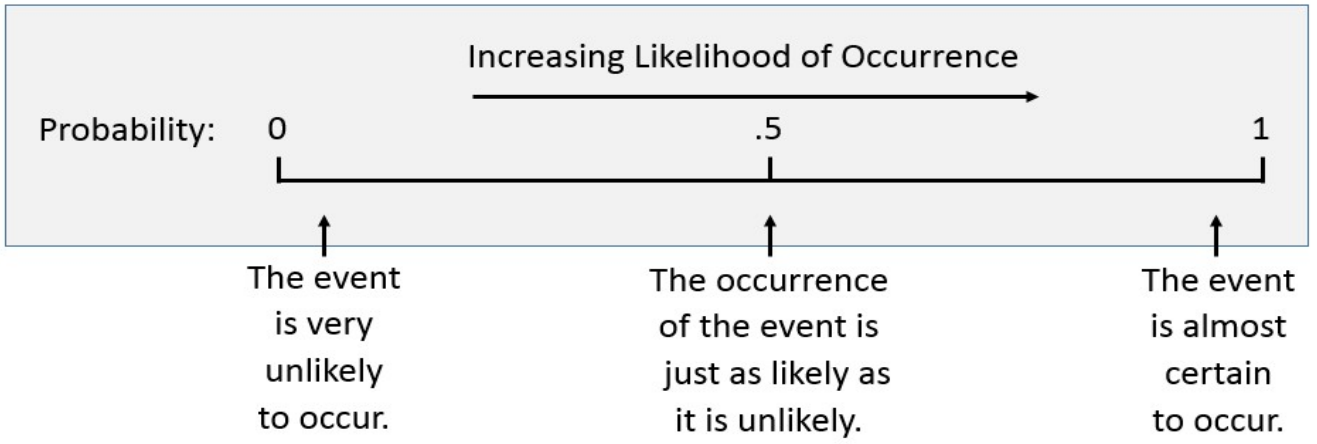
- Managers often base their decisions on an analysis of uncertainties such as the following:
  - What are the chances that the sales will decrease if we increase prices?
  - What is the likelihood a new assembly method will increase productivity?
  - What are the odds that a new investment will be profitable?



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Probability

- Probability is a numerical measure of the likelihood that an event will occur.
- Probability values are always assigned on a scale from 0 to 1.
- A probability near zero indicates an event is quite unlikely to occur.
- A probability near one indicates an event is almost certain to occur.



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Statistical Experiments

- In statistics, the notion of an experiment differs somewhat from that of an experiment in the physical sciences.
- In statistical experiments, probability determines outcomes.
- Even though the experiment is repeated exactly the same way, an entirely different outcome may occur.
- For this reason, statistical experiments are sometimes called *random experiments*.



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Random Experiment and Its Sample Space

- A Random experiment is a process that generates well-defined experimental outcomes.
- The sample space for an experiment is the set of all experimental outcomes.
- An experimental outcome is also called a sample point.

Experiment	Experimental Outcomes
Toss a coin	Head, tail
Inspect a part	Defective, non-defective
Conduct a sale call	Purchase, no purchase
Roll a die	1, 2, 3, 4, 5, 6
Play a football game	Win, loss, tie

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Random Experiment and Its Sample Space

Example: Bradley Investments

Bradley has invested in two stocks, Markley Oil and Collins Mining. Bradley has determined that the possible outcomes of these investments three months from now are as follows:

Investment Gain or Loss in 3 Months (in $1000s):

Markey Oil	Collins Mining
10	8
5	-2
0	
-20	

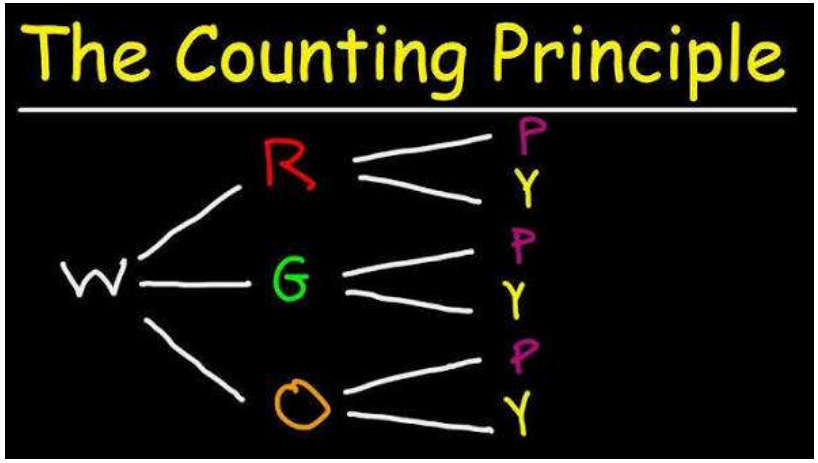
Sample Space

Sample Point

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A Counting Rule for Multiple-Step Experiments

- If an experiment consists of a sequence of k steps in which there are n_1 possible results for the first step, n_2 possible results for the second step, and so on, then the total number of experimental outcomes is given by $(n_1)(n_2) \ldots (n_k)$.
- A helpful graphical representation of a multiple-step experiment is a tree diagram.



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A Counting Rule for Multiple-Step Experiments

Markley Oil: $n_1 = 4$

Collins Mining: $n_2 = 2$

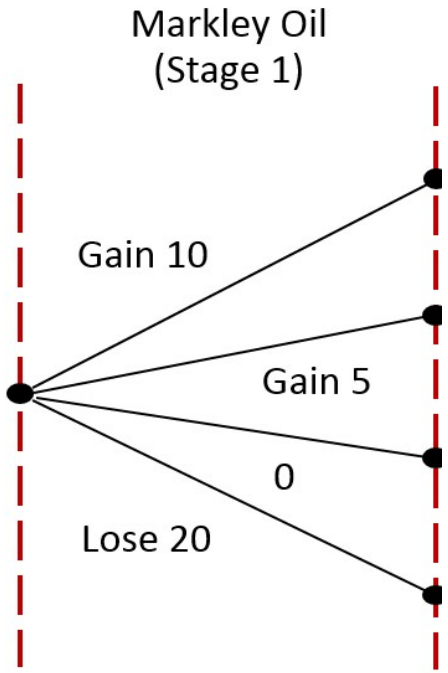
Markey Oil	Collins Mining
10	8
5	-2
0	
-20	

Total number of experimental outcomes: (4)(2) = 8.

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Tree Diagram (1 of 2)

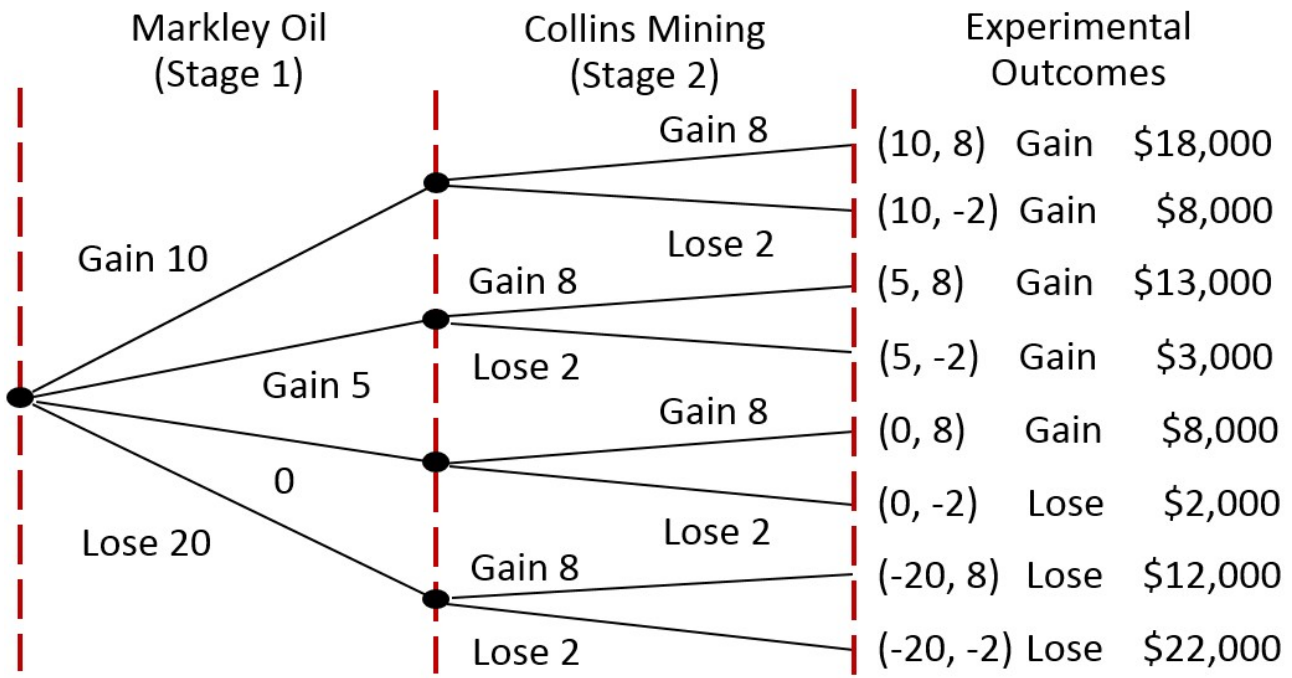
Example: Bradley Investments



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Tree Diagram (1 of 2)

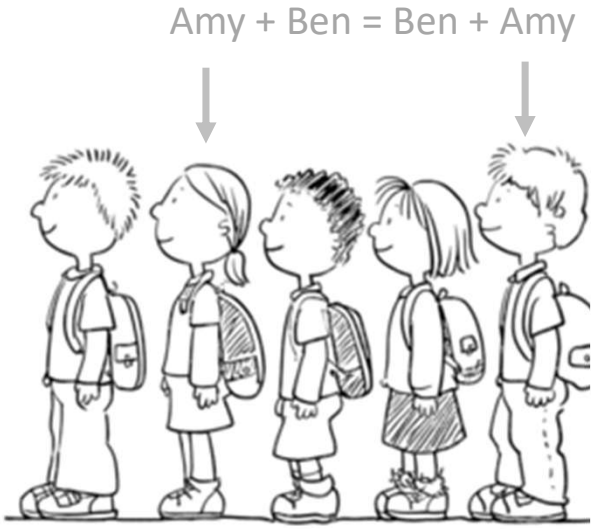
Example: Bradley Investments



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Counting Rule for Combinations

- Number of Combinations of *N* Objects Taken *n* at a Time
- A second useful counting rule enables us to count the number of experimental outcomes when *n* objects are to be selected from a set of *N* objects.
- The order does NOT matter.



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Counting Rule for Combinations

- Number of Combinations of *N* Objects Taken *n* at a Time

$$C_n^N = \binom{N}{n} = \frac{N!}{n!(N-n)!}$$

where: $N! = N(N-1)(N-2)\ldots(2)(1)$

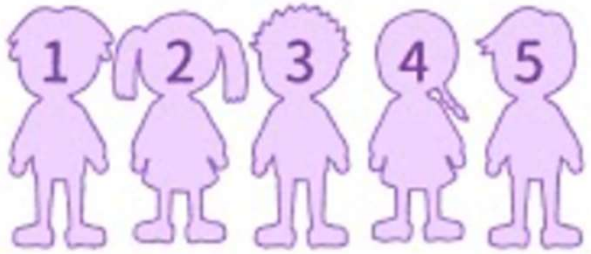
$n! = n(n-1)(n-2)\ldots(2)(1)$

$0! = 1$

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Counting Rule for Permutations

- Number of Permutations of *N* Objects Taken *n* at a Time
- A third useful counting rule enables us to count the number of experimental outcomes when *n* objects are to be selected from a set of *N* objects, where the order of selection is important.
- The order DOES matter.



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Counting Rule for Permutations

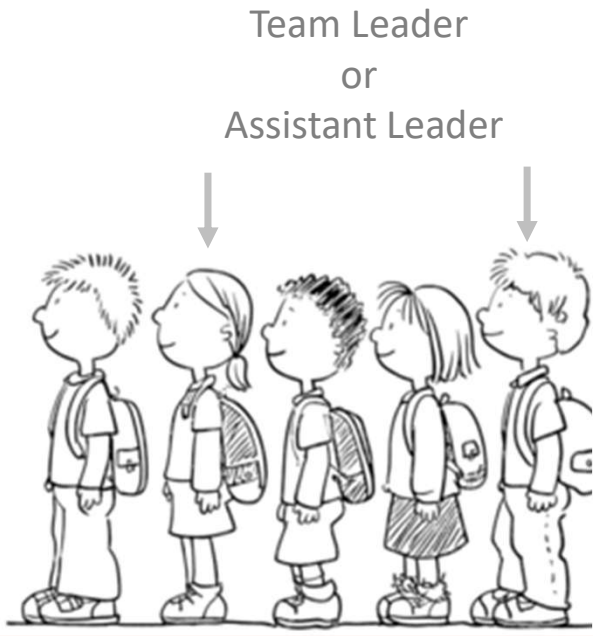
- Number of Permutations of *N* Objects Taken *n* at a Time

$$P_n^N = n!\binom{N}{n} = \frac{N!}{(N-n)!}$$

where: $N! = N(N-1)(N-2)\ldots(2)(1)$

$n! = n(n-1)(n-2)\ldots(2)(1)$

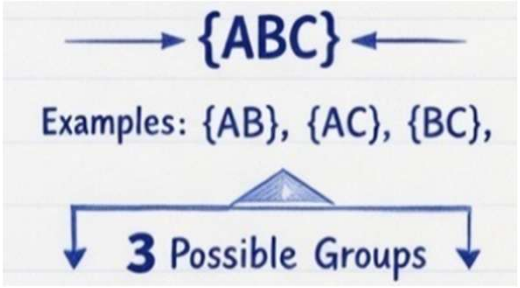
$0! = 1$



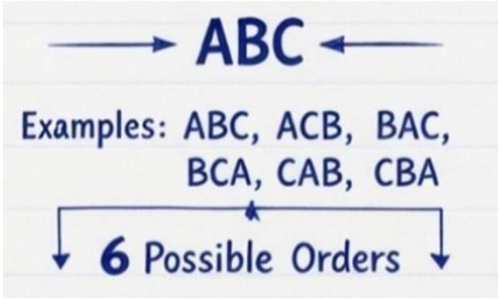
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Combinations VS Permutations









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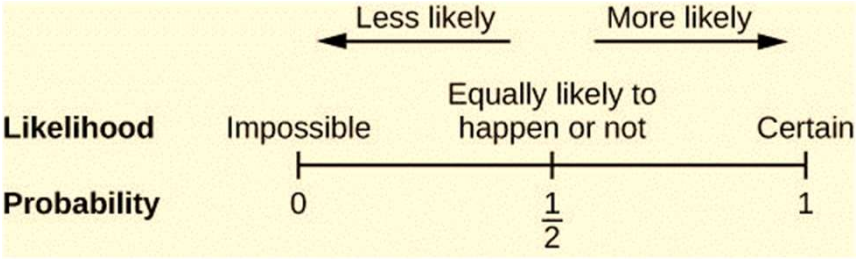
Assigning Probabilities

Basic Requirements for Assigning Probabilities

1. The probability assigned to each experimental outcome must be between 0 and 1, inclusively.

$$0 \leq P(E_i) \leq 1 \text{ for all } i$$

where E_i is the ith experimental outcome and $P(E_i)$ is its probability



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Assigning Probabilities

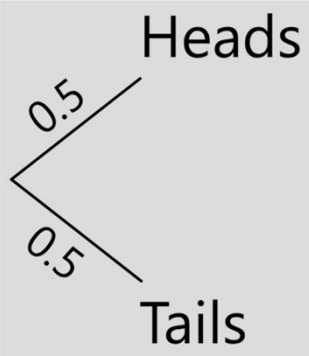
Basic Requirements for Assigning Probabilities

2. The sum of the probabilities for all experimental outcomes must equal 1.

$$P(E_1) + P(E_2) + \ldots + P(E_n) = 1$$

where n is the number of experimental outcomes.





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Assigning Probabilities

- Classical Method

  Assigning probabilities based on the assumption of equally likely outcomes

- Relative Frequency Method

  Assigning probabilities based on experimental or historical data

- Subjective Method

  Assigning probability based on judgment

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Classical Method

Example: Rolling a Die

If an experiment has *n* possible outcomes, the classical method would assign a probability of 1/*n* to each outcome.

Experiment: Rolling a die

Sample Space: $S = \{1, 2, 3, 4, 5, 6\}$

Probabilities: Each sample point has a 1/6 chance of occurring

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Relative Frequency Method

$$Probability = \frac{Frequency\ of\ the\ outcome}{Total\ number\ of\ observations}$$

Example: Lucas Tool Rental

Lucas Tool Rental would like to assign probabilities to the number of car polishers it rents each day. Office records show the following frequencies of daily rental for the last 40 days. Each probability assignment is given by dividing the frequency (number of days) by the total frequency (total number of days).

Number of Polishers Rented	Number of Days	Probability
0	4	
1	6	
2	18	
3	10	
4	2	
sum	40	

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Number of Polishers Rented	Number of Days	Probability
0	4	.10
1	6	.15
2	18	.45
3	10	.25
4	2	.05= 2/40
sum	40	1.00

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Subjective Method

- When economic conditions or a company's circumstances change rapidly, it might be inappropriate to assign probabilities based solely on historical data.



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Subjective Method

- We can use any data available as well as our experience and intuition, but ultimately a probability value should express our degree of belief that the experimental outcome will occur.
- The best probability estimates often are obtained by combining the estimates from the classical or relative frequency approach with the subjective estimate.

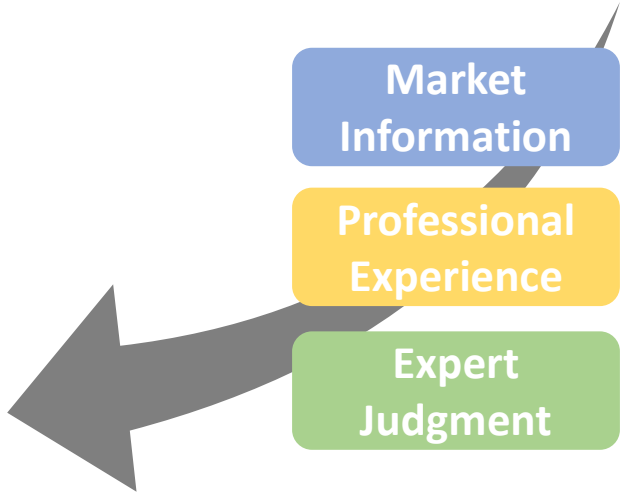
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Subjective Method

Example: Bradley Investments

An analyst made the following probability estimates.

Experimental Outcome	Net Gain or Loss	Probability
(10,8)	$18,000 Gain	
(10, –2)	$8,000 Gain	
(5, 8)	$13,000 Gain	
(5, –2)	$3,000 Gain	
(0, 8)	$8,000 Gain	
(0, –2)	$2,000 Loss	
(–20, 8)	$12,000 Loss	
(–20, –2)	$22,000 Loss	



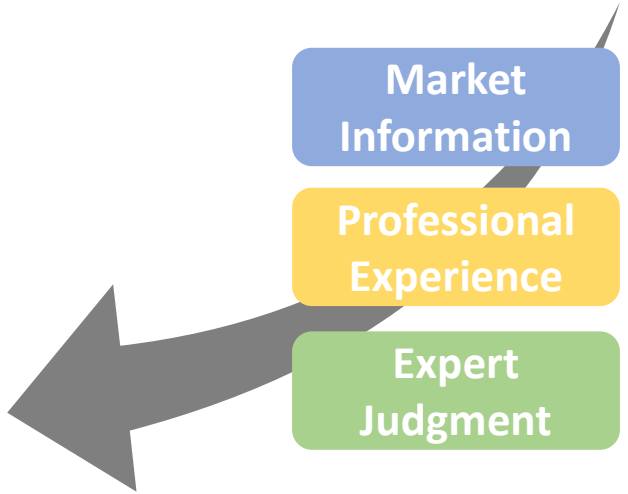
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Subjective Method

Example: Bradley Investments

An analyst made the following probability estimates.

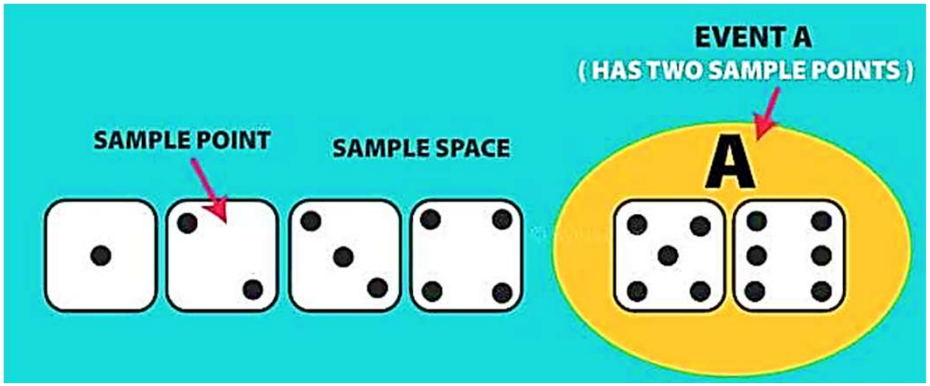
Experimental Outcome	Net Gain or Loss	Probability
(10,8)	$18,000 Gain	0.20
(10, –2)	$8,000 Gain	0.08
(5, 8)	$13,000 Gain	0.16
(5, –2)	$3,000 Gain	0.26
(0, 8)	$8,000 Gain	0.10
(0, –2)	$2,000 Loss	0.12
(–20, 8)	$12,000 Loss	0.02
(–20, –2)	$22,000 Loss	0.06
		Sum equals 1.00



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Events and Their Probabilities

- An event is a collection of sample points.
- The probability of any event is equal to the sum of the probabilities of the sample points in the event.
- If we can identify all the sample points of an experiment and assign a probability to each, we can compute the probability of an event.



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Events and Their Probabilities

Example: Bradley Investments

Markey Oil	Collins Mining
10	8
5	-2
0	
-20	

Event M = Markley Oil is Profitable

$$M = \{ (10, 8), (10, -2), (5, 8), (5, -2) \}$$

$$P(M) = P(10, 8) + P(10, -2) + P(5, 8) + P(5, -2)$$

$$= 0.20 + 0.08 + 0.16 + 0.26$$

$$= 0.70$$

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Events and Their Probabilities

Example: Bradley Investments

Markey Oil	Collins Mining
10	8
5	-2
0	
-20	

Event C = Collins Mining is Profitable

$$C = \{ (10, 8), (5, 8), (0, 8), (-20, 8) \}$$

$$P(C) = P(10, 8) + P(5, 8) + P(0, 8) + P(-20, 8)$$

$$= 0.20 + 0.16 + 0.10 + 0.02$$

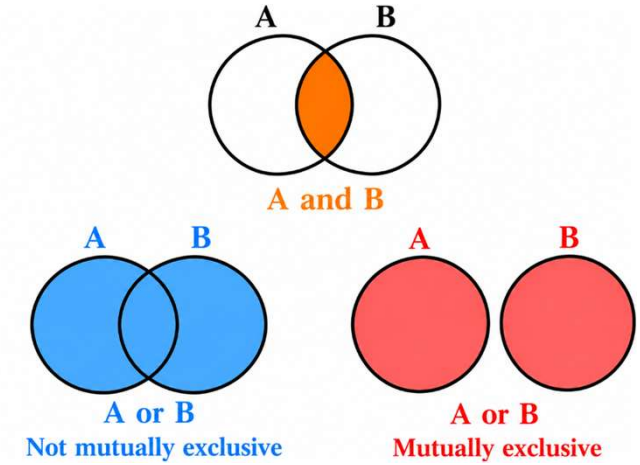
$$= 0.48$$

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Some Basic Relationships of Probability

There are some basic probability relationships that can be used to compute the probability of an event without knowledge of all the sample point probabilities.

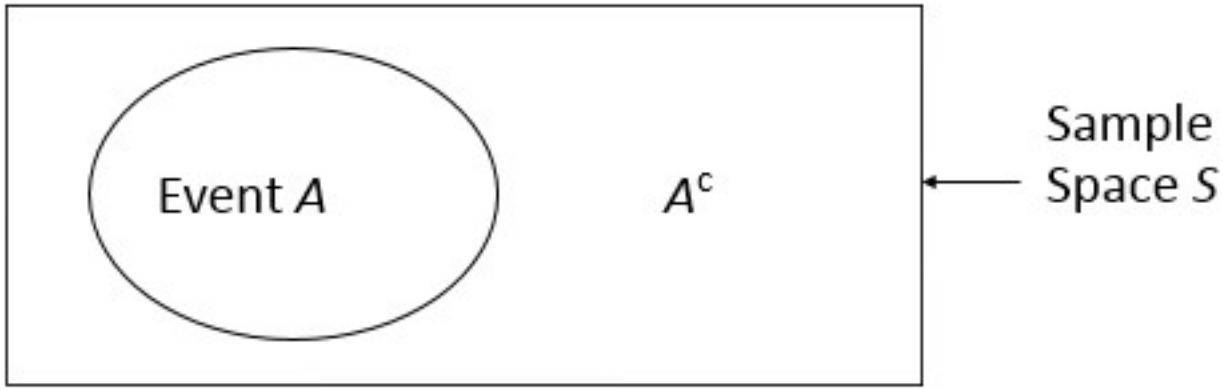
- ✓Complement of an Event (事件的補集)
- ✓Union of Two Events (兩個事件的聯集)
- ✓Intersection of Two Events (兩個事件的交集)
- ✓Mutually Exclusive Events (互斥事件)



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Complement of an Event

- The complement of event A is defined to be the event consisting of all sample points that are *not* in A.
- The complement of *A* is denoted by A^C.

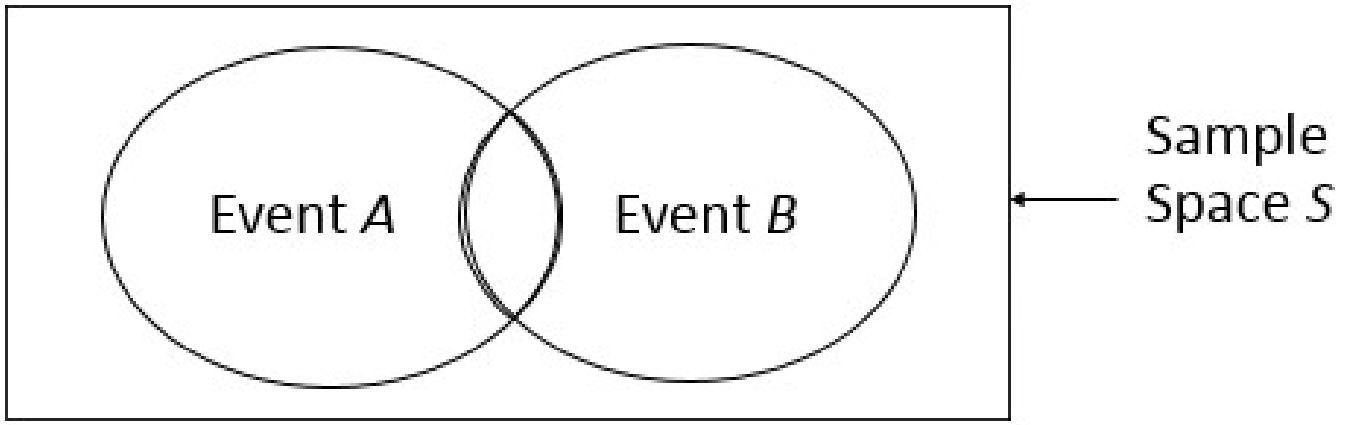


Venn Diagram

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Union of Two Events

The union of events A and B is the event containing all sample points that are in A and B or both. The union of events A and B is denoted by A ∪ B.



Venn Diagram

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Union of Two Events

Example: Bradley Investments

Event M = Markley Oil is Profitable

Event C = Collins Mining is Profitable

$M \cup C$ = Markley Oil is Profitable OR Collins Mining is Profitable (or both are profitable).

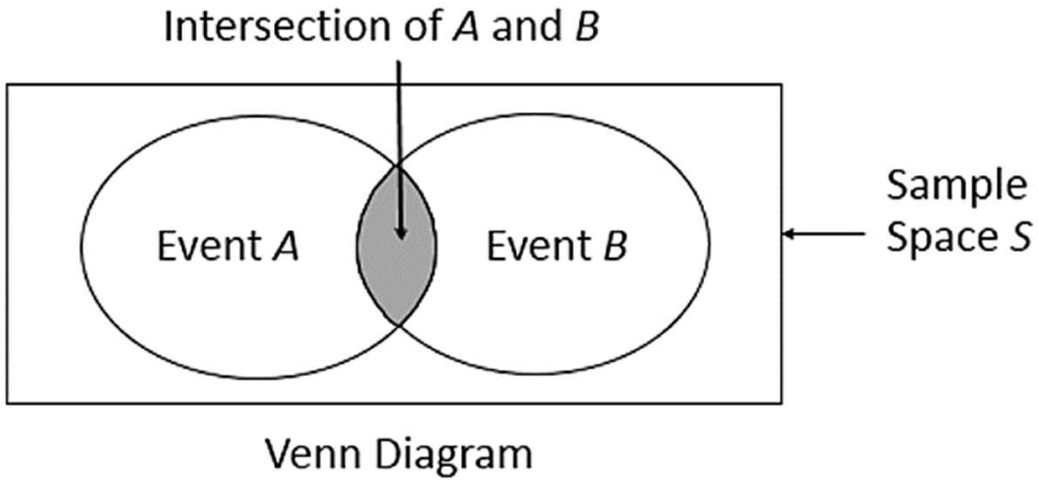
$M \cup C = \{ (10, 8), (10, -2), (5, 8), (5, -2), (0, 8), (-20, 8) \}$

$$P(M \cup C) = P(10, 8) + P(10, -2) + P(5, 8) + P(5, -2) + P(0, 8) + P(-20, 8)$$
$$= 0.20 + 0.08 + 0.16 + 0.26 + 0.10 + 0.02$$
$$= 0.82$$

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Intersection of Two Events

- The intersection of events A and B is the set of all sample points that are in both *A* and *B*
- The intersection of events A and B is denoted by $A \cap B$.



Venn Diagram

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Intersection of Two Events

Example: Bradley Investments

Event M = Markley Oil is Profitable

Event C = Collins Mining is Profitable

$M \cap C$ = Markley Oil is Profitable AND Collins Mining is Profitable.

$M \cap C = \{ (10, 8), (5, 8) \}$

$P(M \cap C) = P(10, 8) + P(5, 8)$

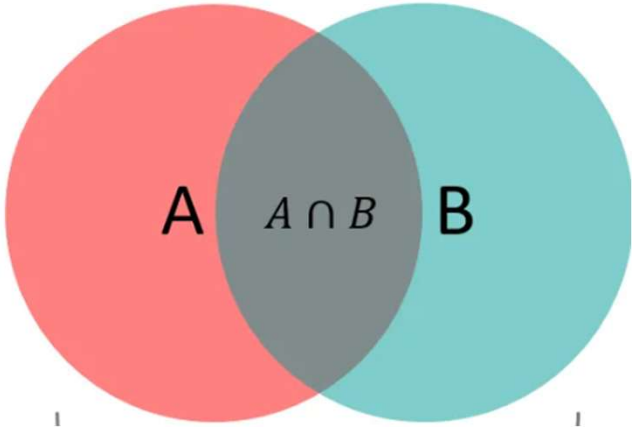
$= 0.20 + 0.16$

$= 0.36$

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Addition Law

- The addition law provides a way to compute the probability of event A, or B, or both *A* and *B* occurring.
- The law is written as: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.



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Addition Law

- The law is written as: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

Event M = Markley Oil is Profitable

Event C = Collins Mining is Profitable

We know that $P(M) = 0.70$, $P(C) = 0.48$, $P(M \cap C) = 0.36$.

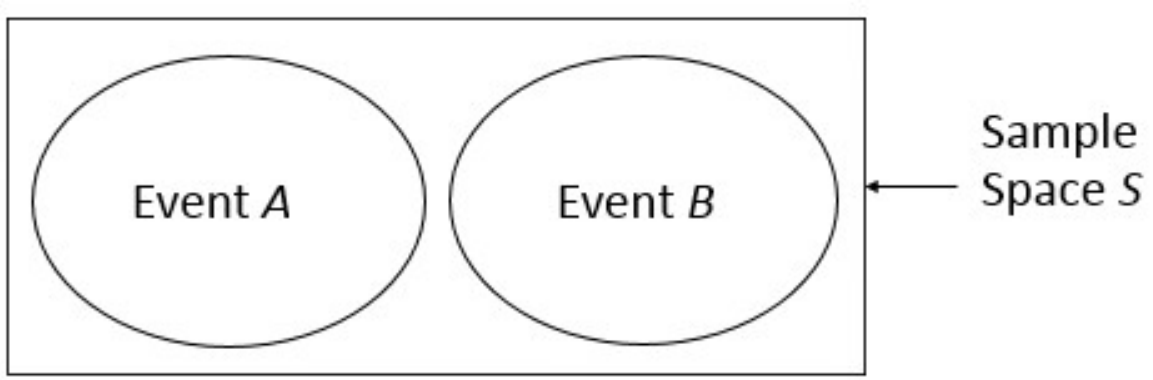
$$\text{Thus } P(M \cup C) = P(M) + P(C) - P(M \cap C)$$
$$= 0.70 + 0.48 - 0.36$$
$$= 0.82$$

Note: This result is the same as that obtained earlier.

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Mutually Exclusive Events

- Two events are said to be mutually exclusive if the events have no sample points in common
- Two events are mutually exclusive if, when one event occurs, the other cannot occur.

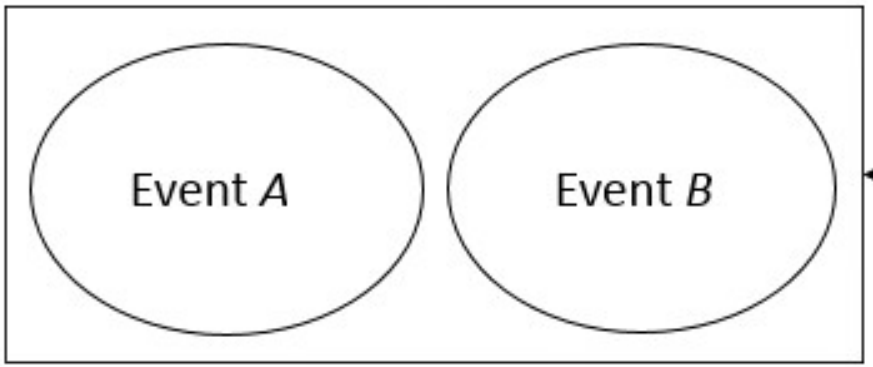


Venn Diagram

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Mutually Exclusive Events

- If events A and B are mutually exclusive, then $P(A \cap B) = 0$.
- The addition law for mutually exclusive events is $P(A \cup B) = P(A) + P(B)$.



Venn Diagram

Addition Law➔

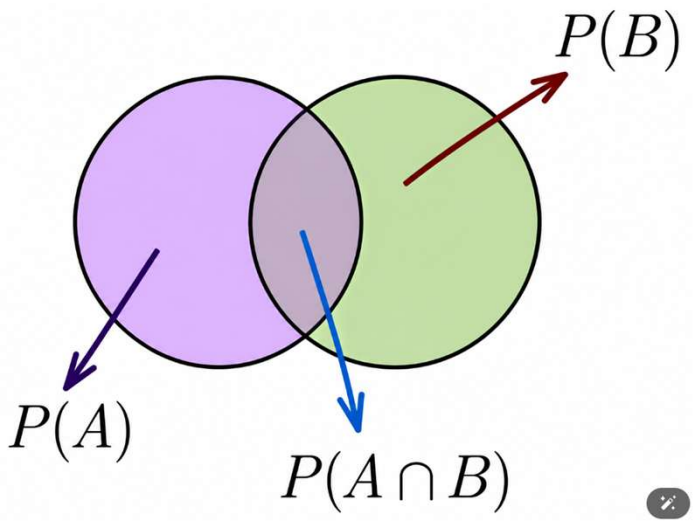
$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Example:
Let event A be: Rolling an even number
And let event B be: Rolling an odd number

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Conditional Probability

- The probability of an event given that another event has occurred is called a conditional probability.
- The conditional probability of *A* given B has already occurred denoted by $P(A \mid B)$.



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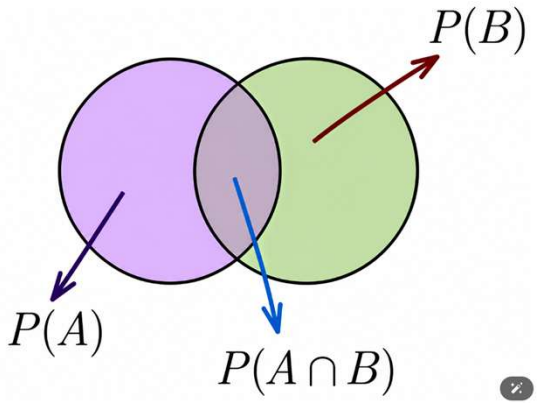
Conditional Probability

- A conditional probability is computed as follows:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

← The probability that both A and B occur

← The probability that B occurs



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Conditional Probability

Example: Bradley Investments

Event M = Markley Oil Profitable

Event C = Collins Mining Profitable

$P(C|M)$ = Collins Mining Profitable given Markley Oil Profitable

We know: $P(M \cap C) = .36$, $P(M) = .70$

Thus: $P(C|M) = \frac{P(C \cap M)}{P(M)} = \frac{.36}{.70} = .5143$

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Multiplication Law

- The multiplication law provides a way to compute the probability of the intersection of two events.

Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Rearrange

Multiplication Law

$$P(A \cap B) = P(B)P(A|B) \quad \text{or} \quad P(A \cap B) = P(A)P(B|A)$$

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Multiplication Law

- The multiplication law provides a way to compute the probability of the intersection of two events.
- The law is written as: $P(A \cap B) = P(B)P(A|B)$ or $P(A \cap B) = P(A)P(B|A)$

Event M = Markley Oil is Profitable

Event C = Collins Mining is Profitable

We know that $P(M) = 0.70$ and $P(C|M) = 0.5143$

$$P(M \cap C) = P(M) \cdot P(C|M)$$
$$= (0.70)(0.5143)$$
$$= 0.36$$

Note: This is the same result we obtained earlier.

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Joint Probability Table (聯合機率表)

- Joint probabilities appear in the body of the table

Markley Oil	Collins mining Profitable (C)	Collins mining not Profitable (C^C)	Total
Profitable (M)	.36	.34	.70
Not Profitable (M^C)	.12	.18	.30
Total	.48	.52	1.00

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Joint Probability Table (聯合機率表)

- Marginal probabilities appear in the margins of the table

Markley Oil	Collins mining Profitable (C)	Collins mining not Profitable (C^C)	Total
Profitable (M)	.36	.34	.70
Not Profitable (M^C)	.12	.18	.30
Total	.48	.52	1.00

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Independent Events

- If the probability of event *A* is not changed by the existence of event *B*, we would say that events *A* and *B* and are independent.
- Two events *A* and *B* are independent if:

$$P(A \mid B) = P(A) \quad \text{or} \quad P(B \mid A) = P(B)$$

Multiplication Rule

Independent Events

$$P(A \text{ and } B) = P(A) \cdot P(B)$$

Dependent Events

$$P(A \text{ and } B) = P(A) \cdot P(B \mid A)$$

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Multiplication Law for Independent Events

- The multiplication law also can be used as a test to see if two events are independent.
- The law is written as: $P(A \cap B) = P(A)P(B)$

Multiplication Rule

Independent Events

$$P(A \text{ and } B) = P(A) \bullet P(B)$$

Dependent Events

$$P(A \text{ and } B) = P(A) \bullet P(B \mid A)$$

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Multiplication Law for Independent Events

$$P(A \cap B) = P(A)P(B)$$

Event M = Markley Oil is Profitable

Event C = Collins Mining is Profitable

Are events M and C independent?

Does $P(M \cap C) = P(M)P(C)$?

We know that $P(M \cap C) = 0.36$, $P(M) = 0.70$ and $P(C) = 0.48$

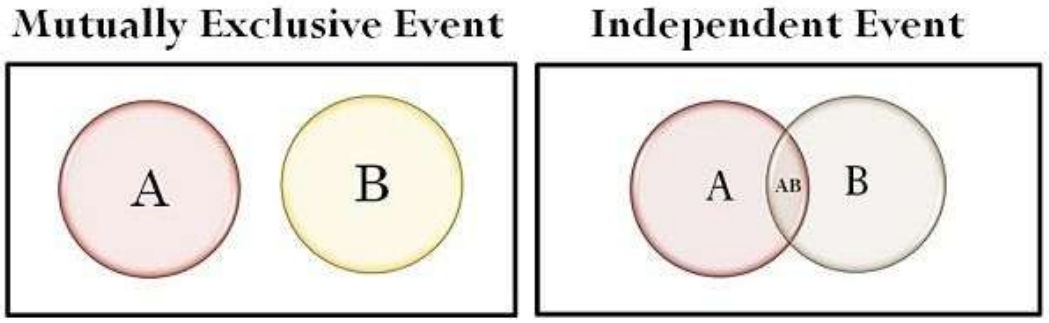
But $P(M)P(C) = (0.7)(0.48) = 0.34$, which does not equal 0.36.

Therefore M and C are not independent.

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Mutual Exclusiveness and Independence

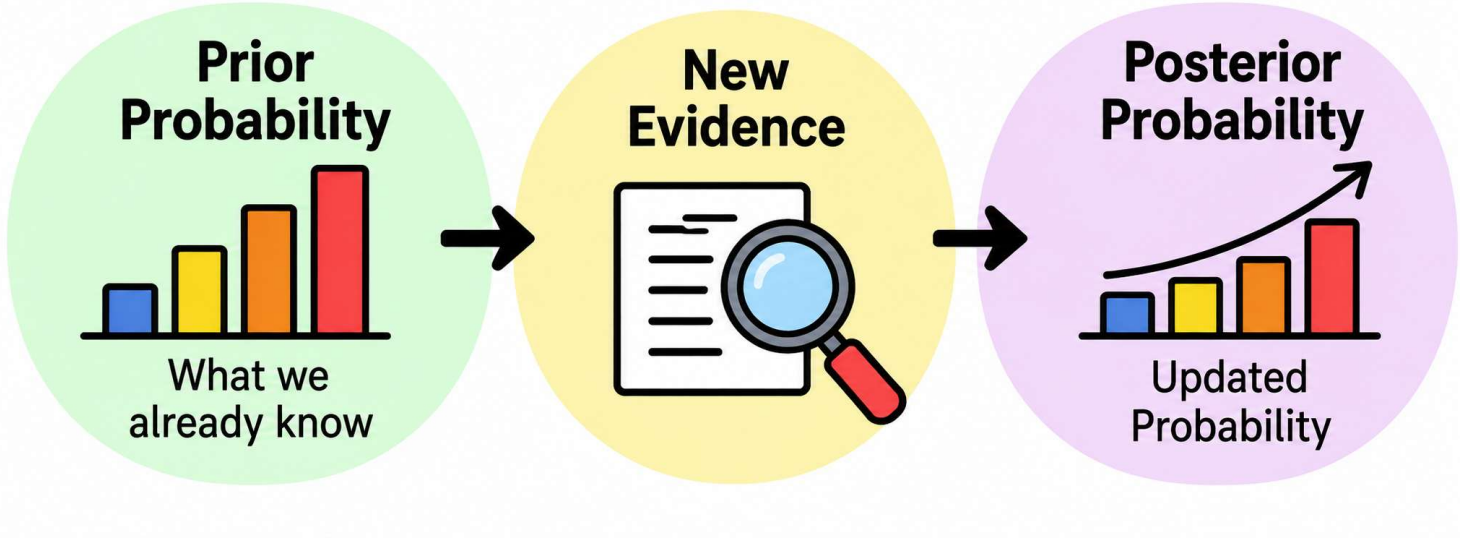
- Do not confuse the notion of mutually exclusive events with that of independent events.
- Two events with nonzero probability cannot both mutually exclusive and independent.
- If one mutually exclusive event is known to occur, the other cannot occur; thus, the probability of the other event occurring is reduced to zero (and therefore dependent).
- Two events that are not mutually exclusive might or might not be independent.



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Bayes' Theorem

- Often we begin probability analysis with initial or prior probabilities (先驗機率).
- Then, from a sample, special report, or a product test, we obtain some additional information.



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Bayes' Theorem

- Given this information, we calculate revised or posterior probabilities (事後機率).
- Bayes' theorem provides the means for revising the prior probabilities.



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Bayes' Theorem

A proposed shopping center will provide strong competition for downtown businesses like L. S. Clothiers. If the shopping center is built, the owner of L. S. Clothiers feels it would be best to relocate to the shopping center.

The shopping center cannot be built unless a zoning change is approved by the town council. The planning board must first make a recommendation, for or against the zoning change, to the council.

Let: A_1 = town council approves the zoning change

A_2 = town council disapproves the change

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Let: A_1 = town council approves the zoning change

A_2 = town council disapproves the change

Using subjective judgment: $P(A_1) = 0.7$ and $P(A_2) = 0.3$

Prior Probabilities

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New Information

The planning board has recommended **against** the zoning change. Let *B* denote the event of a negative recommendation by the planning board.

Given that *B* has occurred, should L. S. Clothiers revise the probabilities that the town council will approve or disapprove the zoning change?

Past history with the planning board and the town council indicates the following: Hence:

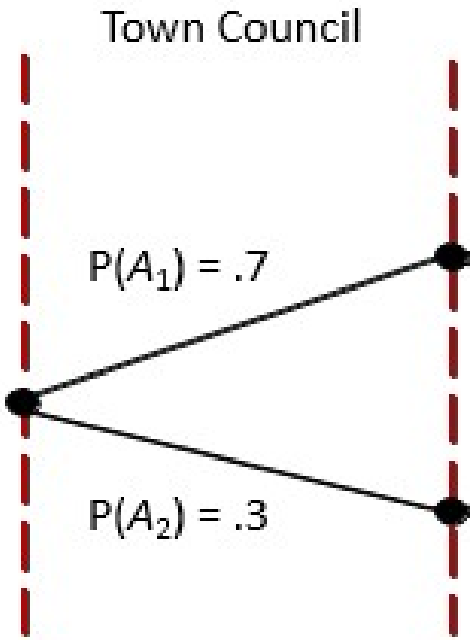
$$P(B \mid A_1) = 0.2 \text{ and } P(B \mid A_2) = 0.9$$

$$P(B^c \mid A_1) = 0.8 \text{ and } P(B^c \mid A_2) = 0.1$$

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Tree Diagram

Example: L. S. Clothiers

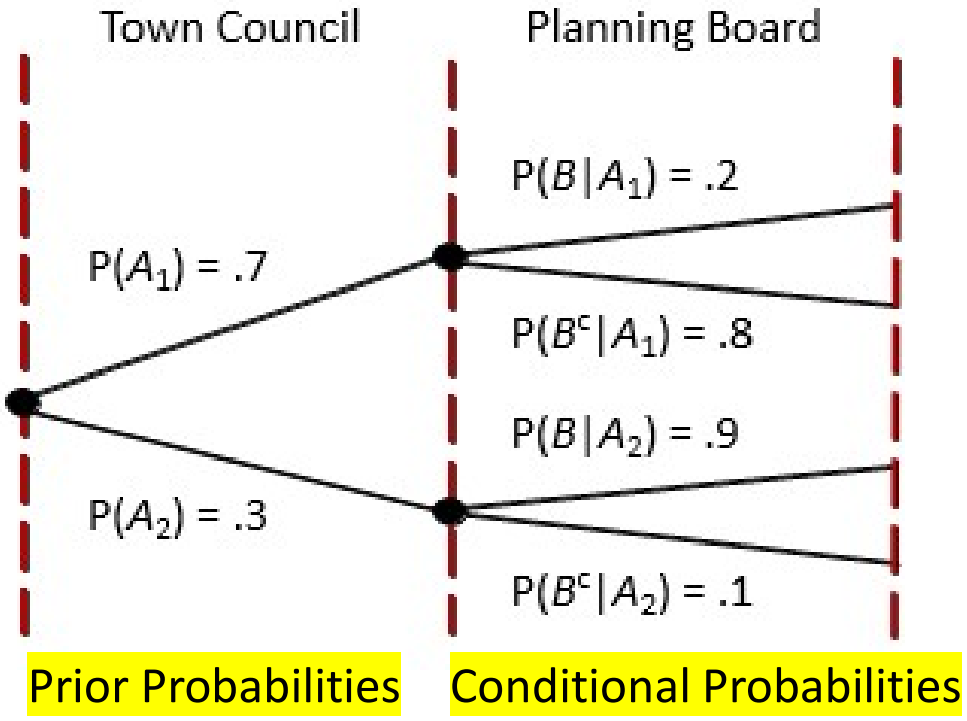


Prior Probabilities

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Tree Diagram

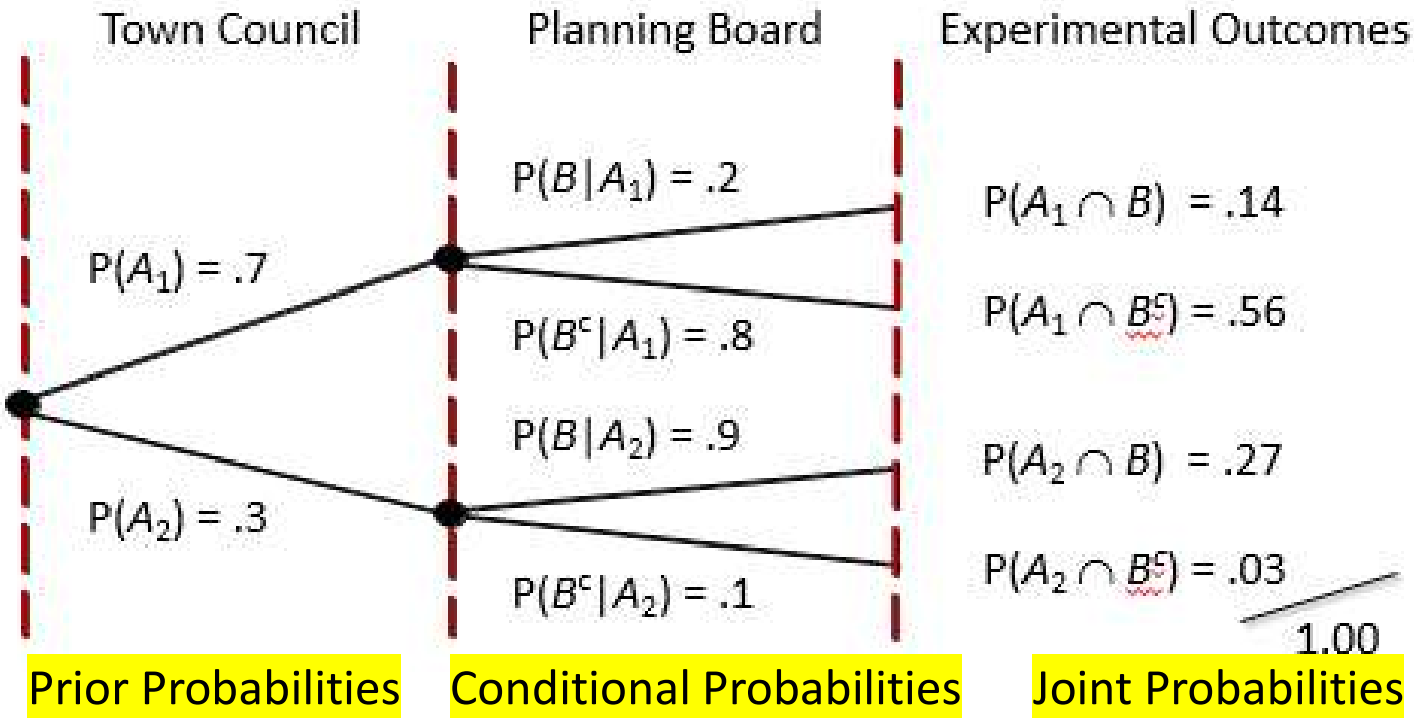
Example: L. S. Clothiers



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Tree Diagram

Example: L. S. Clothiers



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Bayes' Theorem

- To find the posterior probability that event A_i will occur given that event B has occurred, we apply Bayes' theorem.

Joint Probabilities

$$P(A_i|B) = \frac{P(A_i)P(B|A_i)}{P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + \cdots + P(A_n)P(B|A_n)}$$

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Bayes' Theorem

- To find the posterior probability that event A_i will occur given that event B has occurred, we apply Bayes' theorem.

$$P(A_i|B) = \frac{P(A_i)P(B|A_i)}{P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + \cdots + P(A_n)P(B|A_n)}$$

The total probability that event B occurs

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Bayes' Theorem

- To find the posterior probability that event A_i will occur given that event B has occurred, we apply Bayes' theorem.

$$P(A_i|B) = \frac{P(A_i)P(B|A_i)}{P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + \cdots + P(A_n)P(B|A_n)}$$

- Bayes' theorem is applicable when the events for which we want to compute posterior probabilities are mutually exclusive and their union is the entire sample space.

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Posterior Probabilities

Example: L. S. Clothiers

Given the planning board's recommendation not to approve the zoning change, we revise the prior probabilities as follows:

$$P(A_1|B) = \frac{P(A_1)P(B|A_1)}{P(A_1)P(B|A_1) + P(A_2)P(B|A_2)}$$

$$= \frac{(0.7)(0.2)}{(0.7)(0.2)+(0.3)(0.9)}$$

$P(A_1 \cap B) = .14$

$P(A_2 \cap B) = .27$

The planning board's recommendation is good news for L. S. Clothiers. The posterior probability of the town council approving the zoning change is 0.34 compared to a prior probability of 0.70.

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Posterior Probabilities

Example: L. S. Clothiers

Given the planning board's recommendation not to approve the zoning change, we revise the prior probabilities as follows:

$$P(A_1|B) = \frac{P(A_1)P(B|A_1)}{P(A_1)P(B|A_1) + P(A_2)P(B|A_2)}$$

$$= \frac{(0.7)(0.2)}{(0.7)(0.2)+(0.3)(0.9)} = 0.34$$

The planning board's recommendation is good news for L. S. Clothiers. The posterior probability of the town council approving the zoning change is 0.34 compared to a prior probability of 0.70.

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Bayes' Theorem: Tabular Approach

Step 1: Prepare the following three columns:

(1) Events A_i	(2)	(3)	(4)	(5)
A_1				
A_2				

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Bayes' Theorem: Tabular Approach

Step 1: Prepare the following three columns:

(1)	(2)	(3)	(4)	(5)
Events	Prior Probabilities			
A_i	$P(A_i)$			
A_1	.7			
A_2	.3			
	1.0			

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Bayes' Theorem: Tabular Approach

Step 1: Prepare the following three columns:

(1) Events A_i	(2) Prior Probabilities $P(A_i)$	(3) Conditional Probability $P(B \mid A_i)$	(4)	(5)
A_1	.7	.2		
A_2	.3	.9		
	1.0			

If A_i happens, what is the probability of observing B?
But not: Given B, what is the probability of A_i?
➔That is what we are eventually trying to find.

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Bayes' Theorem: Tabular Approach

Step 2: Prepare the fourth column

Column 4: Compute the joint probabilities for each event and the new information *B* by using the multiplication law. Multiply the probabilities in column 2 by the corresponding conditional probabilities in column 3.

(1)	(2)	(3)	(4)	(5)
Events	Prior Probabilities	Conditional Probability	Joint Probabilities	
A_i	$P(A_i)$	$P(B \mid A_i)$	$P(A_i \cap B)$	
A_1	.7	.2		
A_2	.3	.9		
	1.0			

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

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Bayes' Theorem: Tabular Approach

Step 2: Prepare the fourth column

Column 4: Compute the joint probabilities for each event and the new information *B* by using the multiplication law. Multiply the probabilities in column 2 by the corresponding conditional probabilities in column 3.

(1) Events A_i	(2) Prior Probabilities $P(A_i)$	(3) Conditional Probability $P(B \mid A_i)$	(4) Joint Probabilities $P(A_i \cap B)$	(5)
A_1	.7	.2	.14 = .7(.2)	
A_2	.3	.9	.27	
	1.0			

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

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Bayes' Theorem: Tabular Approach

Step 3

Sum the joint probabilities in Column 4. The sum is the probability of the new information, $P(B)$. The sum 0.14 + 0.27 shows an overall probability of 0.41 of a negative recommendation by the planning board.

(1)	(2)	(3)	(4)	(5)
Events	Prior Probabilities	Conditional Probability	Joint Probabilities	
A_i	$P(A_i)$	$P(B \mid A_i)$	$P(A_i \cap B)$	
A_1	.7	.2	.14 = .7 (.2)	
A_2	.3	.9	.27	
	1.0		P(B) = .41	

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Bayes' Theorem: Tabular Approach

Step 4

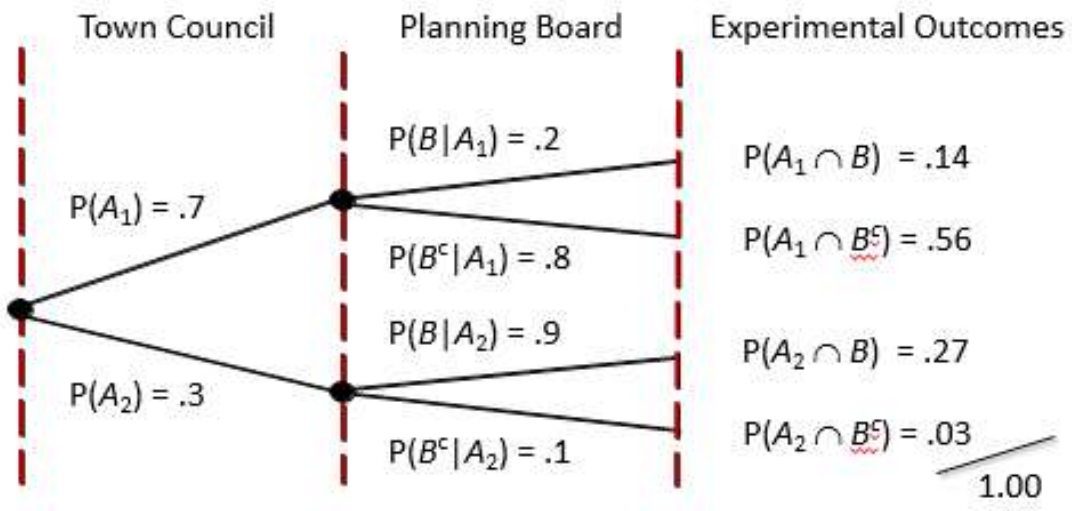
Prepare the fifth column: Column 5

Compute the posterior probabilities using the basic relationship of conditional probability.

(1) Events A_i	(2) Prior Probabilities $P(A_i)$	(3) Conditional Probability $P(B \mid A_i)$	(4) Joint Probabilities $P(A_i \cap B)$	(5) Posterior Probabilities $P(A_i \mid B)$
A_1	.7	.2	.14 = .7(.2)	.3415 = .14/.41
A_2	.3	.9	.27	.6585
	1.0		P(B) = .41	1.0000

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Bayes' Theorem



Use new information to update probabilities

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$



(1) Events A_i	(2) Prior Probabilities $P(A_i)$	(3) Conditional Probability $P(B\|A_i)$	(4) Joint Probabilities $P(A_i \cap B)$	(5) Posterior Probabilities $P(A_i\|B)$
A_1	.7	.2	.14 = .7(.2)	.3415 = .14/.41
A_2	.3	.9	.27	.6585
	1.0		P(B) = .41	1.0000

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